

Analysis of X-ray Spectra by Iterative Least Squares (AXIL): New Developments*

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The functionality of the computer package AXIL (Analysis of X-ray spectra by Iterative Least squares), suitable for the evaluation of energy-dispersive X-ray spectra, has been extended in a number of ways. First, a background modelling algorithm, based on the use of mutually orthogonal polynomials was introduced to replace the linear or exponential polynomials employed previously. Second, the Gaussian photopeak model employed in previous versions of the program was extended to include the non-Gaussian parts of the characteristic peaks and related background contributions. Both innovations are shown to improve the performance of the spectrum evaluation procedure. Third, the user-friendliness of the evaluation program was enhanced (a) by allowing a PC plug-in MCA card to be directly controlled from within the program and (b) by extending the command-interface to allow for repeated execution of series of commands by means of control loops. A brief description of these changes is given; as an application, the accurate evaluation of micro-XRF images is discussed.

INTRODUCTION

Energy-dispersive X-ray spectrometry (EDXRS), using X-rays, electrons or protons to induce the emission of characteristic radiation from a material, is a widely established technique of elemental analysis. Although the energy-dispersive method of detection is mechanically more simple and cost effective than the wavelength-dispersive alternative, the relatively low-energy resolution of Si(Li) detectors (in the range 150–200 eV) complicates the evaluation of ED-XRS spectra considerably. Simple approaches to net peak determination, which yield acceptable results in the case of wavelength-dispersive data, do not produce reliable or accurate results in the case of ED spectra.

In our laboratory, from 1977 onwards, efforts were made to develop a dedicated computer code for the evaluation (sometimes also called deconvolution) of complex energy-dispersive X-ray spectra.¹ The resulting program package is called AXIL (Analysis of X-ray spectra by Iterative Least squares) and is based on the non-linear fitting of a mathematical function (describing the photopeaks and the spectral background) to experimental data. The original versions of AXIL dealt exclusively with spectrum evaluation *per se* and were implemented on PDP-11 computers.^{1–3} In 1985, a PC version of the program was implemented, featuring the possibility of interactively building and testing spectrum models.⁴ Together with the actual fitting program, accessory programs to specify the general parameters of the fit model and to access or edit a library of X-ray line energies ε_{jk} and relative line ratios R_{jk} were made avail-

able. Also, a number of utility programs were provided to convert spectrum files in the format produced by popular MCAs (multi-channel analysers) to a standard format or to read out these MCAs using a PC. During the period 1985–90, a number of calibration models of increasing complexity for quantitative ED-XRF analysis were implemented and incorporated into the AXIL package, allowing users to convert net peak intensities, which were obtained during the spectrum evaluation step, into meaningful chemical concentrations.^{5,6} An overview of the various modules of the package and their interdependences is shown in Fig. 1.

In this paper, a number of extensions to the spectrum evaluation module of the AXIL package are described. Most of these innovations were prompted by the need to be able to use the spectrum evaluation procedure in an increasingly unsupervised manner, without having to compromise on the accuracy of the net peak area determination. An area where unsupervised processing of large series of ED-XRF spectra is very useful is micro-XRF (μ -XRF). In this microscopical variant of bulk ED-XRF, a microscopic X-ray beam is used to excite locally a small area (e.g. $10 \times 10 \mu\text{m}^2$) of the sample surface. By moving the sample through the microbeam and collecting an ED-XRF spectrum from each location, X-ray maps of the elemental constituents of the material under study can be collected. However, even the collection of fairly small images (e.g. 50×50 pixels) implies the processing of large number of spectra (2500 in this example). Quasi-unsupervised spectrum evaluation requires procedures which are more robust than the algorithms that are interactively used.

In order to increase the robustness and accuracy of the spectrum evaluation procedure, a new background compensation method which employs a linear combination of mutually orthogonal polynomials as a background fitting model was incorporated. In addition, the purely Gaussian photopeak shape, which was used in

* Presented at the European Conference on EDXRS, Budapest, Hungary, May 30–June 3, 1994.

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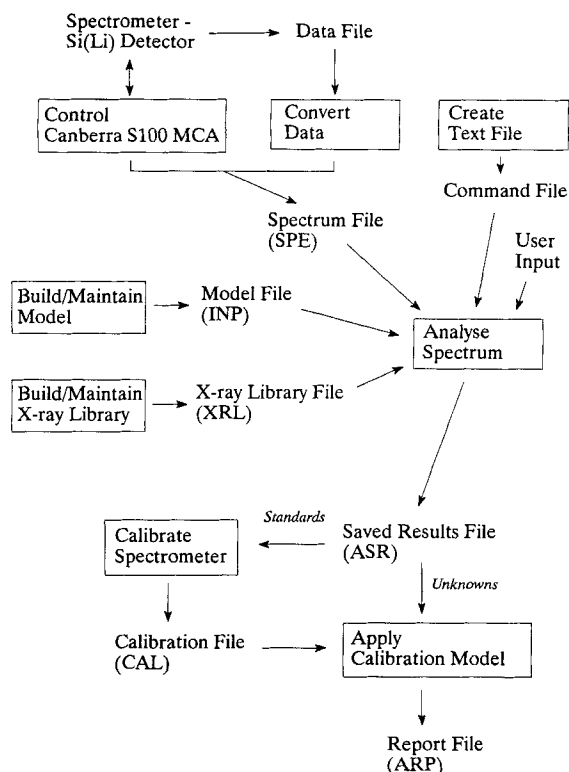


Figure 1. Overview of the various modules and their interdependencies in the software package AXIL.

previous versions of the program, was extended to include non-Gaussian tails and background contribution associated with each photopeak.

Having the use of the package for μ -XRF image acquisition and processing in mind, two other changes were introduced. In order to arrive at an elemental XRF image, a series of spectra, each representing a pixel in the image, must be processed. Usually, the same series of commands must be executed for each of the (many) spectra constituting such an image. Thus, the command interface of the spectrum evaluation program was enhanced to allow for the repeated execution of a number of steps in the spectrum evaluation job by means of loop-control commands. The addition of a module through which Canberra S100 PC plug-in MCA cards can be controlled by the fitting program, allowing data acquisition and processing to be done in parallel, together with the above-mentioned batchwise processing ability, automates the collection and generation of elemental XRF images in an efficient way.

MATHEMATICAL BACKGROUND

Non-linear least-squares fitting

In AXIL, the spectrum evaluation is based on building a mathematical model with which the analyst tries to describe the experimental data. This is done by specifying the region of the spectrum to be fitted [the region-of-interest (ROI)], the selection of a suitable background compensation method, the selection of a

number of X-ray line groups (e.g. Fe K, Ba L_{III}, Pb L_I) to be included in the model, and by establishing approximately correct values for the energy and resolution parameters of the spectrometer. The model parameters are then optimized by means of a non-linear least-squares strategy, using a modified Marquardt algorithm⁷ to minimize the weighted sum of differences χ^2 between the experimental data y_i and the mathematical model function $y_{\text{model}}(i)$:

$$\chi^2 = \frac{1}{n - m} \sum_i \frac{[y_i - y_{\text{model}}(i)]^2}{y_i} \quad (1)$$

where n is the total number of channels in the ROI and m represents the number of parameters to be optimized during the fitting process. The model parameters can be divided into four major groups: (a) peak group areas, (b) calibration parameters (energy and resolution), (c) background model parameters (if any) and (d) peak shape parameters (see below). The model function $y_{\text{model}}(i)$ may consist of two terms describing the spectral background and the photopeaks:

$$y_{\text{model}}(i) = y_{\text{back}}(i) + \sum_j A_j \sum_{k=1}^{N_j} R_{jk} y_f(i, \epsilon_{jk}) \quad (2)$$

where the index j runs over all characteristic line groups which appear in the spectrum. The lines in group j (N_j in total) are modelled by means of Gaussian functions $y_f(i, \epsilon_{jk})$. During the fitting process, the algorithm attempts to find the minimum of the χ^2 surface in the multidimensional parameter space. For the purpose of spectrum evaluation, the final values of the group area parameters A_j (or rather the peak areas $R_{jk} A_j$) are of primary interest.

Background compensation methods

Recently, a number of background compensation methods have been proposed⁸ and in order to evaluate the performance of these methods, they were incorporated in the spectrum evaluation module of AXIL. A distinction must be made between two types of background compensation strategies. The first type are the *background estimation* methods. Here, the background shape $b(i)$ is estimated (by applying a peak-stripping or filtering algorithm to the experimental data y_i) and subtracted from the original spectrum y_i ; the resulting net spectrum $n_i = y_i - b(i)$ is then further evaluated. The second type are the *background modelling* methods, which attempt to adjust the parameters of a suitable mathematical model $y_{\text{back}}(i)$ to the background shape during the fitting process simultaneously with all the other parameters of $y_{\text{model}}(i)$ [Eqn (2)]. In what follows, a brief description of the various background compensation methods considered here is given.

Background estimation methods. The SNIP method. As proposed by Clayton and co-workers,^{9,10} this method, called Statistical Non-linear Iterative Peak clipping (SNIP), is based on the comparison of the content y_i of channel i with the mean m_i of the contents of neighbouring channels a distance w away from channel i . If m_i is smaller than y_i , the content of this channel is

replaced by the mean m_i :

$$m_i = \frac{y_{i-w} + y_{i+w}}{2}$$

IF $y_i > m_i$ THEN $y_i = m_i$ (3)

The width w is initially set to twice the peak-full-width-at-half-maximum $\text{FWHM}(i)$. This conditional substitution procedure is executed for all channels in the ROI. Repeating this procedure N times (typically 10–20 times) gradually causes peaks to be ‘stripped’ away from the original spectral data, so that only the estimate of the background shape $b(i)$ remains.

The ORTPOL method. An alternative peak-stripping algorithm was proposed by Steenstrup et al.¹¹ This method describes the background estimate $b(i)$ as a linear combination of mutually orthogonal polynomials $P_l(i)$:

$$b(i) = \sum_{l=0}^L a_l P_l(i) \quad (4)$$

where L denotes the degree of the polynomial. The orthogonal polynomials $P_l(i)$ are constructed by means of the following recurrence relationship:

$$\begin{aligned} P_0(i) &= 1 \\ P_1(i) &= i - \alpha_0 \\ P_l(i) &= (i - \alpha_{l-1})P_{l-1}(i) - \beta_{l-1}P_{l-2}(i) \\ &\quad (l = 2, \dots, L-1) \end{aligned}$$

where

$$\begin{aligned} \alpha_0 &= \frac{\sum_{i=1}^n w(i)i}{\sum_{i=1}^n w(i)}, \quad \alpha_l = \frac{\sum_{i=1}^n w(i)i[P_l(i)]^2}{\gamma_l} \\ \beta_0 &= 0, \quad \beta_l = \frac{\sum_{i=1}^n w(i)i[P_l(i)P_{l-1}(i)]}{\gamma_{l-1}} \quad (l = 1, \dots, L) \end{aligned} \quad (5)$$

The values of the coefficients a_l can simply be calculated by the weighted sum of the polynomials:

$$\begin{aligned} a_l &= \sum_i \frac{w(i)y_i P_l(i)}{\gamma_l} \\ \gamma_l &= \sum_i w(i)[P_l(i)]^2 \end{aligned} \quad (6)$$

where $w(i)$ represents the weight of channel i and is equal to $1/b(i)$ when channel i belongs to a background region of the spectrum; when channel i is part of a peak, the weight is set to 0. Since prior to the calculation the background estimate is not known, all channel weights are initially set to $1/y_i$. After calculating a first estimate of the coefficients a_l , the channel weights are adjusted and the entire procedure repeated. Channel i is identified as being part of a peak when y_i is found to be significantly higher than the current estimate $b(i)$. This procedure is repeated a number of times until a stable set of weights $w(i)$ is obtained.

Background modelling methods. *The EXPON method.* As part of the fitting function of Eqn (2), an exponential polynomial can be used to model the background:

$$y_{\text{back}}(i) = \exp \left[\sum_{l=0}^L b_l (i - i_0)^l \right] \quad (7)$$

Because the b_l are non-linear fitting parameters, nearly correct initial values for these parameters need to be provided. In order to achieve a converging background fit, this is done by estimating the background $b(i)$ by means of the SNIP algorithm (see above), and performing a linear fit of the logarithm of the obtained background to a polynomial $\sum b_l (i - i_0)^l$. The exponential background modelling method has been incorporated from the very first versions of the AXIL program¹ and is often applied to fit the background of an entire spectrum or of spectral regions with high positive or negative curvature. The order L of the polynomial may vary between 1 and 8.

The LINEAR method. In previous versions of the AXIL program, it was also possible to employ an ordinary linear polynomial to model the background:

$$y_{\text{back}}(i) = \sum_{l=0}^L a_l (i - i_0)^l \quad (8)$$

Because the a_l are linear fitting parameters, they do not need to be initialized, so the use of this method was more straightforward. However, the major disadvantage of this method is that because of fitting singularities, the maximum order L should not exceed 4 and, as a result, the linear model of Eqn (8) can only be employed to describe the background of relatively small spectral regions (< 3 keV) or to model relatively simple background shapes. In the present version of the program, this simple model has been replaced by a linear combination of mutually orthogonal polynomials:

$$y_{\text{back}}(i) = \sum_{l=0}^L a_l P_l(i) \quad (9)$$

As a first step the orthogonal polynomials $P_l(i)$ are determined using the ORTPOL method. During the fitting process the background coefficients a_l are adjusted together with the other parameters in the fitting model. Because of the orthogonality of the polynomials $P_l(i)$, the order L can be as high as 30 without causing singularity problems during the fitting process. Thus, the function can be used to model more complex background shapes and/or the background of an entire spectrum.

RESULTS AND DISCUSSION

Evaluation of the various background compensation methods

Figure 2 shows the scheme that was used to evaluate the four above described background compensation methods. Simulated data which very closely resemble experimental spectra [see Fig. 3(a) and Fig. 5(a)], but

Mean Squared Error (MSE) of estimator

$$= (\text{variance of estimator}) + (\text{its bias})^2$$

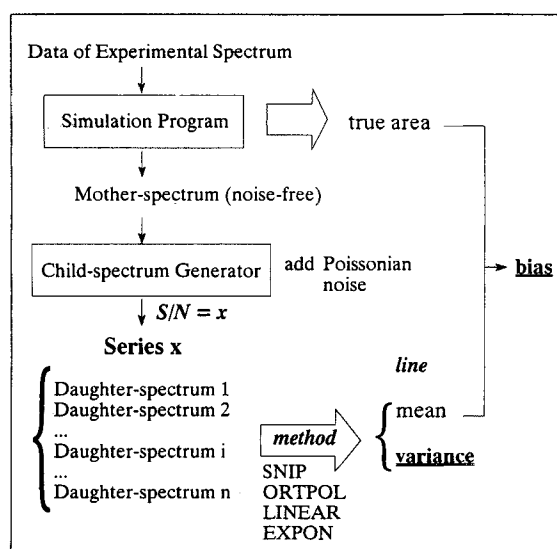


Figure 2. Scheme of the procedure for evaluation of the various background compensation methods.

which are noise-free and in which the peak intensities are *a priori* known, are used as so-called 'mother-spectra'^{12,13} Using an algorithm suggested by Ryan *et al.*,¹⁰ on the basis of these mother distributions, series of daughter spectra are generated, each with a specific distribution of noise counts. By fitting various series of these daughter spectra, each having a different total intensity $T = \sum y_i$ and hence characterized by a different overall signal-to-noise ratio equal to $T/\sqrt{T}$, the accuracy and precision with which the four methods are able to compensate for the background and, thus, their ability to aid in determining the net peak areas in the spectra via non-linear least-squares fitting can be evaluated. Details of this evaluation procedure can be found in Ref. 14. The bias (i.e. the difference between the mean of the net peak area estimates and the true peak area) and variance (the degree to which peak area estimates tend to spread about the mean) of each method are combined into a single figure-of-merit called the mean squared error (MSE)¹⁵:

$$\text{MSE} = \text{bias}^2 + \text{variance} \quad (10)$$

Details on the calculation of the bias and the variance are given in Ref. 14 and Fig.2. The corresponding relative quantity is called the efficiency E_{jk} of the background compensation method k for estimating the true area A_j of the j^{th} peak (group) in the spectrum:

$$E_{jk} = \frac{\sqrt{\text{MSE}_{jk}}}{A_j} \times 100\% \quad (11)$$

As an example, in Fig. 3(b), the root-mean-squared efficiency value $E_{\text{rms}, k}$:

$$E_{\text{rms}, k}(\%) = \sqrt{\frac{\sum_j E_{jk}^2}{J}} \quad (12)$$

where J is the number of lines in the spectrum, is plotted for the four methods for the case of the NIST SRM 1571 Orchard Leaves spectrum shown in Fig. 3(a). Figure 3(c) and (d) illustrate the performance of the various background compensation methods when one of the daughter spectra is fitted; all methods are able to produce background shapes which, at least visually, closely resemble what one intuitively assumes to be the background shape. Figure 3(b) shows that the LINEAR method is superior to the other methods. A more elaborate study¹⁴ concludes that the method in which the background is modelled as a linear combination of orthogonal polynomials is identified as being the most robust and yielding the most accurate results. However, even for the series with the lowest noise-to-signal ratio, there still remains a relative bias of approximately 2%, indicating that the applied model [Eqn (2)] is still not completely able to describe the spectral data. One of the reasons that the background functions systematically under- or overestimate the net peak areas is that Gaussian peak shapes are used as peak models.

Non-Gaussian peak shapes

Corrections for the tailing of intense peaks are necessary to accurately estimate the net peak areas of small peaks in the neighbourhood of intense lines. To correct for these artifacts in the spectra, peak shape corrections (PSC) can be applied in the AXIL spectrum evaluation code. These are numerical corrections which, after appropriate scaling, may be added to a Gaussian peak group $y_j(i)$ and can correct for the presence of non-Gaussian tails of intense photopeaks. In principle, these numerical values need to be determined for each detector separately, using experimental spectra with very good counting statistics. The application of PSC improves the performance of the fitting methods significantly.^{2,14} Another way to deal with the (low-energy) tails is to model them by means of an appropriate analytical function. This can be done in a number of ways by means of the Hypermet function.¹⁶⁻¹⁸ In this work, the Hypermet function, consisting of a Gaussian function $G(i)$ augmented with an exponential tail $D(i)$ and a flat shelf $S(i)$ (Fig. 4), was used:

$$y_j(i) = G(i) + S(i) + D(i)$$

$$G(i) = H_G \exp \left[\frac{-(i - i_0)^2}{2\sigma^2} \right]$$

$$S(i) = \frac{H_S}{2} \operatorname{erfc} \left(\frac{i - i_0}{\sigma\sqrt{2}} \right)$$

$$D(i) = \frac{H_D}{2} \exp \left(\frac{i - i_0}{\beta} \right) \operatorname{erfc} \left(\frac{i - i_0}{\sigma\sqrt{2}} + \frac{\sigma}{\beta\sqrt{2}} \right) \quad (13)$$

where H_G , H_S and H_D are the height of the Gaussian, the flat shelf and the exponential tail, respectively. The parameter β gives the slope of the exponential tail and σ is the width of the Gaussian. In the fitting model, the peak shape parameters H_S , H_D and β are expressed as

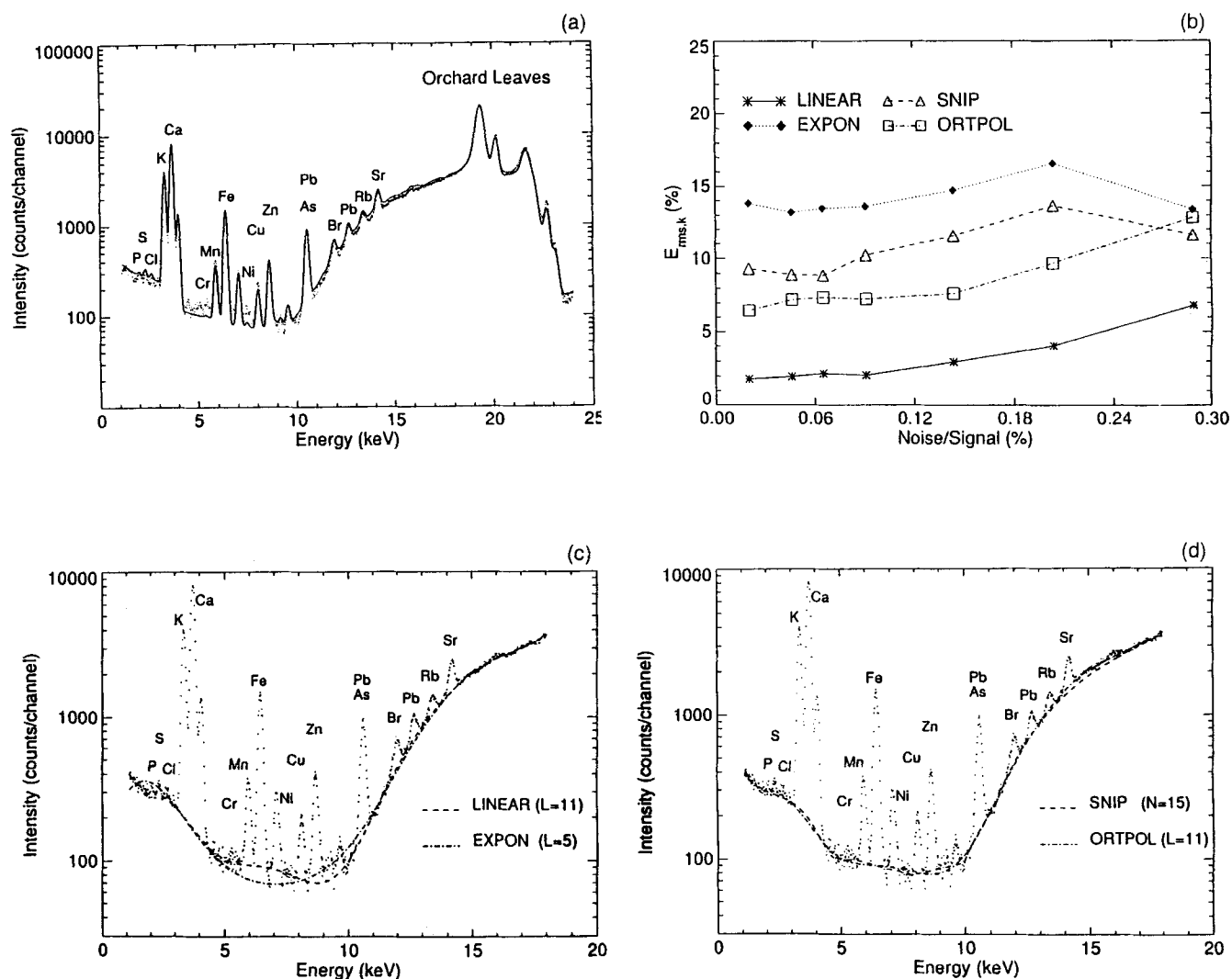


Figure 3. Results of the evaluation of the various background compensation methods when processing series of daughter spectra of NIST SRM 1571 Orchard Leaves: (a) experimental (dots) and simulated mother (solid lines) spectra; (b) $E_{ms,k}$ as a function of noise-to-signal ratio when series of daughter spectra are processed with optimal parameter values; (c and d) performance of the four background compensation methods when a daughter spectrum (noise-to-signal = 0.06%) is fitted with optimum parameter values.

function of the photopeak energy, as suggested by Campbell *et al.*,¹⁷ in a similar way to the spectrometer resolution and energy calibration parameters. In this work, no attempt was made to describe the radiative

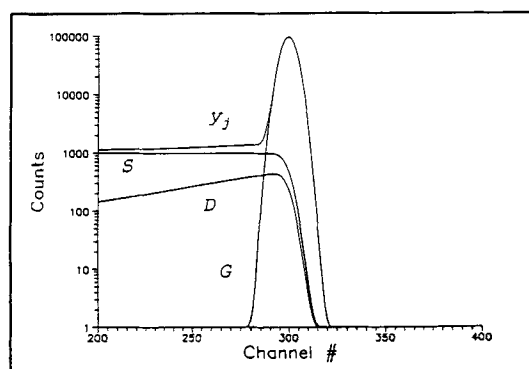


Figure 4. Hypermet function to describe the peak shape of an x-ray line. The step function S , the experimental tail D and the Gaussian function G are shown, together with the envelope y_j .

Auger contributions which augment the tailing of the $K\alpha$ and $K\beta$ lines with separate analytical functions, i.e. the KMM radiative Auger feature which is situated in the tail of a $K\beta$ peak is not separately accounted for. Hence, when the Hypermet function is employed to describe the Si(Li) spectral lineshapes, the applied background function has to account for the deviations between the experimental data and the applied line-shape functions. In this respect, only background modelling procedures can be used in an effective way together with these non-Gaussian peak shapes. Figure 5(b)–(d) shows the efficiency E_{jk} for the minor lines (Ti, Mn, Co and Cu) when a series of daughter spectra with a noise-to-signal ratio of 0.13% (signal-to-noise ratio = 769) of a NIST SRM 1208-1 Inco 718 Steel spectrum [see Fig. 5(a)] is evaluated with the LINEAR background compensation method, as a function of the order of the background polynomial employed for three cases: using the Gaussian peak model with and without PSC [Fig. 5(b) and (c), respectively] and the Hypermet peak model [Fig. 5(d)]. In the spectrum in Fig. 5(a) a

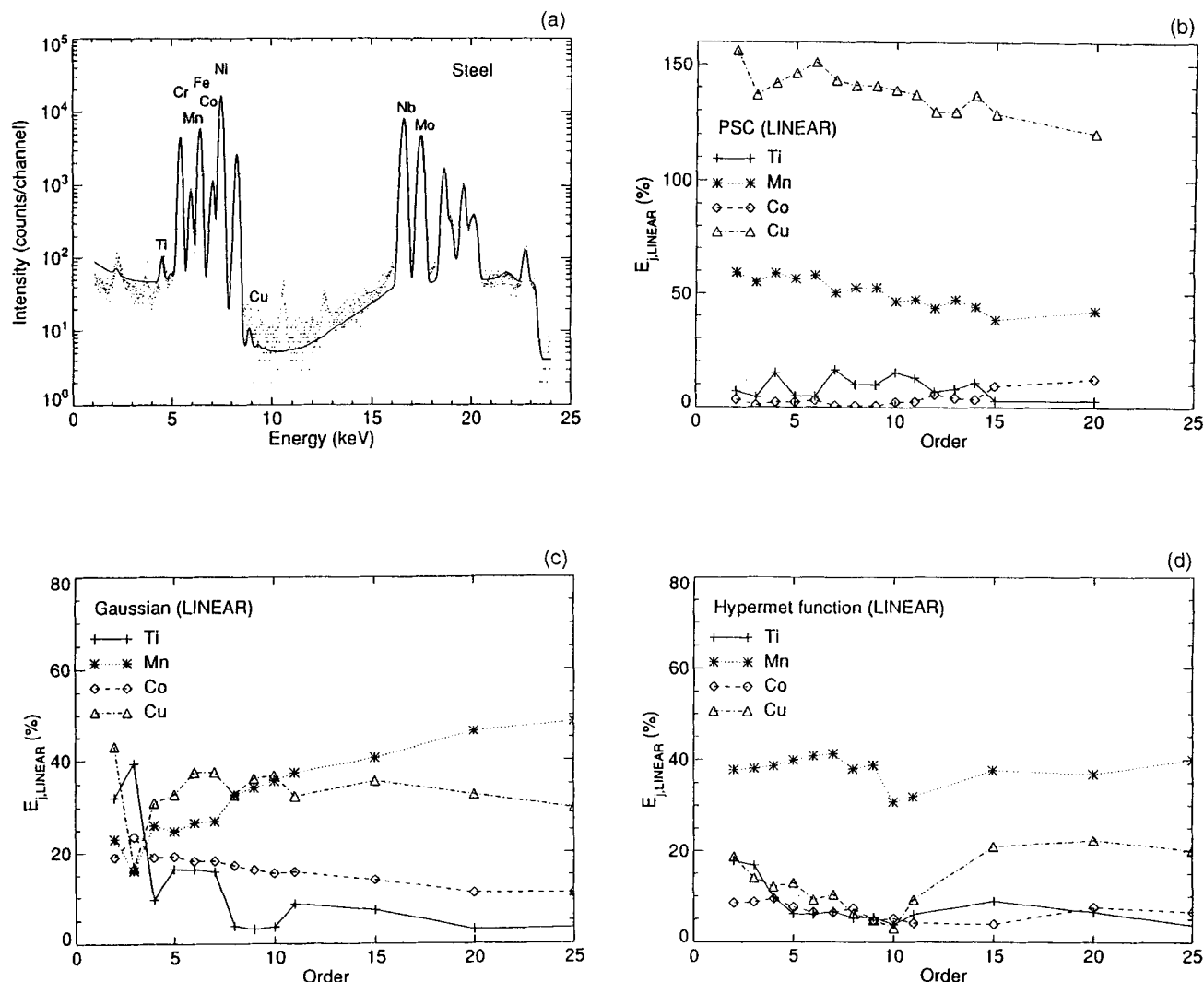


Figure 5. Evaluation of three different models using the LINEAR method to estimate net peak areas of small peaks (Ti, Mn, Co, Cu) in the neighbourhood of intense lines (Cr, Fe, Ni) when processing a series of daughter spectra (noise-to-signal ratio = 0.13% or signal-to-noise ratio = 769) of the NIST SRM 1208-1 Inco 718 Steel spectrum: (a) experimental (dots) and simulated mother (solid lines) spectra; $E_{rms,k}$ as a function of the degree of the polynomial for (b) the Gaussian peak model with PSC, (c) Gaussian peak model without PSC and (d) the Hypermet peak model.

double difficulty occurs: on the one hand severe overlap of Cr $K\beta$ /Mn $K\alpha$, Fe $K\beta$ /Co $K\alpha$ and Ni $K\beta$ /Cu $K\alpha$, and on the other hand the presence of the non-Gaussian tails of the Cr $K\alpha$, Fe $K\alpha$ and Ni $K\alpha$ lines. The overlap makes it difficult to determine precisely the $K\alpha$ intensities of Mn, Co and Cu, while self-absorption in the sample and the non-Gaussian components of the matrix lines affect the accuracy with which the net intensities of Ti $K\alpha$, Mn $K\alpha$, Co $K\alpha$ and Cu $K\alpha$ can be determined. As shown in Fig. 5(d), the modelling of the non-Gaussian tails improves the performance of the LINEAR method. The efficiency for most of the minor lines is approximately 5–10% for polynomial orders of 5–10 when modelling the tails, whereas without any corrections, mean errors of 5–40% are obtained. Note that the net peak area of Mn $K\alpha$ cannot be estimated efficiently because the small Mn $K\alpha$ peak nearly coincides with the intense line of Cr $K\beta$. A major disadvantage of employing PSC is that these numerical

corrections are scaled to the $K\alpha$ intensity of the peak group to which they belong, resulting in this case in an underestimation of the Ni $K\beta$ line together with a relatively serious and systematic underestimation of the much smaller Cu $K\alpha$ intensity. Similar results are obtained for other types of spectra in which intense matrix lines appear close to smaller peaks. A more elaborate comparison between the use of the Hypermet function as photopeak model and the use of PSC with Gaussian peak shapes is given elsewhere.¹⁹

X-ray image processing

To accommodate for the evaluation of large numbers of spectra, the possibility of using extensive loop-control commands was introduced into the command language of the AXIL spectrum evaluation module. An example

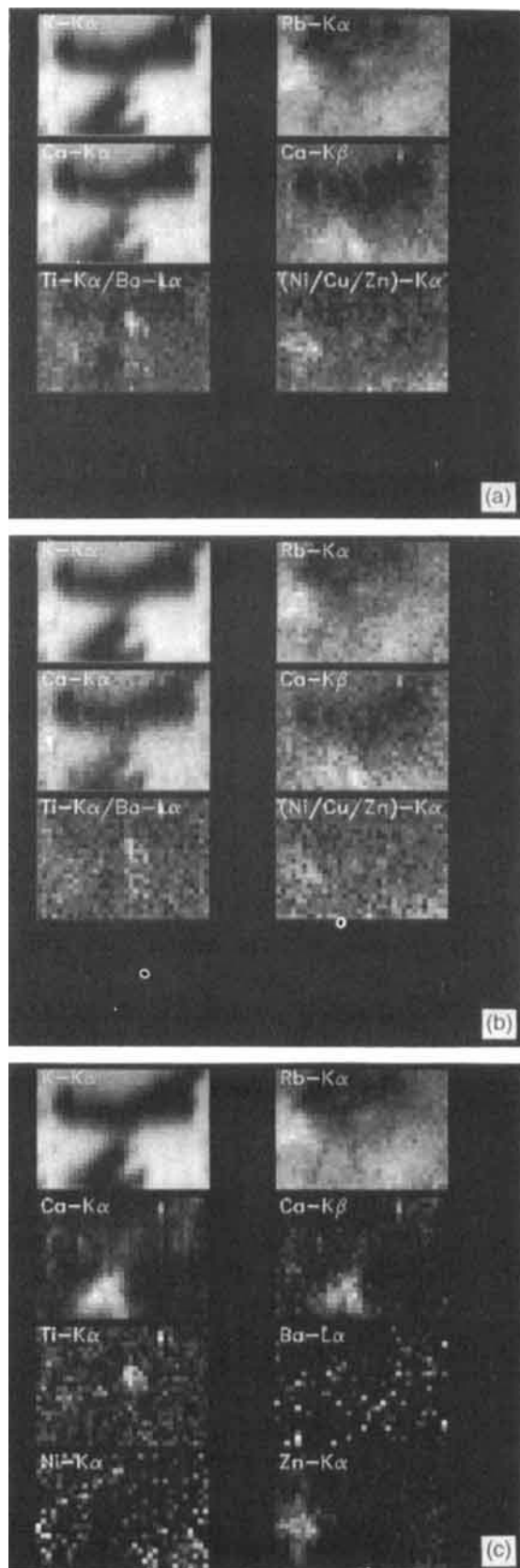


Figure 6. X-ray images resulting from processing 24×34 X-ray image spectra, collected from a geological two-phase feldspar system (microcline perthite), using three different methods: (a) method A sums the contents of channels within a number of ROIs; (b) method B corrects for the background by linear interpolation between the edges of the ROIs; (c) method C uses spectrum evaluation software (AXIL).

of a typical command sequence to fit automatically a series of spectra (SPEC1.SPE, SPEC2.SPE, ..., SPEC816.SPE) can be as follows:

```
LOAD MODEL=IMAGMOD.INP
WITH $1 FROM 1 TO 816
LOAD SPEC=SPEC$1
FIT N_ITER=50
SAVE_RES
ENDWITH
```

Here, after loading the appropriate fitting model (IMAGMOD.INP), all spectra of the series are read from disk and fitted with a maximum of 50 iterations, after which the resulting net peak intensities are saved for later processing.

The present version of the program also contains a module through which the user can simultaneously control up to four CANBERRA S100 MCA cards. From within the fitting program itself, it is now possible to perform functions such as: select one of the cards, select the memory region of the chosen card, clear the selected memory region, send a signal to the sample changer, set real or life time, start the acquisition, monitor the growing spectrum and store the acquired spectrum on a disk. A recently acquired spectrum can also be evaluated while the next spectrum is being acquired. Also, by means of the above-mentioned loop commands and the built-in sample changer control facility, large numbers of spectra (originating from one or more samples) can be acquired automatically.

To illustrate the necessity of employing accurate spectrum evaluation when processing X-ray image data, a set of 24×34 ED-XRF spectra, collected from a geological two-phase feldspar system (microcline perthite) using an $8 \times 8 \mu\text{m}^2$ polychromatic synchrotron microbeam, were processed in three different ways. Method A [Fig. 6(a)] simply involved the summing of channels within a number of ROIs, without any background correction. Method B [Fig. 6(b)] is similar to A, but attempts to correct for the background by linear interpolation between the edges of the ROI, taking the mean of three channel contents left and right of the edge as reference points. In method C [Fig. 6(c)], a least-squares spectrum evaluation was performed on each spectrum in the image using the AXIL software. Whereas the images of the major lines (e.g. K $K\alpha$ and Rb $K\alpha$) hardly depend on the spectrum evaluation method being used, the resolving power of AXIL and its (more reliable) background compensation ability are necessary to accurately determine the spatial distribution of elements such as Ca and Zn. Also, in contrast with the methods A and B, for overlapping trace elements lines (e.g. Ni $K\alpha$ and Cu $K\alpha$, Ti $K\alpha$ and Ba $L\alpha$) the deconvolution is much easier when using spectrum evaluation software, resulting in more accurate X-ray images with higher contrast.

CONCLUSIONS

Recently, a number of improvements to the spectrum evaluation program of the AXIL software package have

been incorporated. One was the introduction of a number of recently proposed background compensation methods. A comparative study shows that the LINEAR method seems to be the optimum choice (considering robustness, processing speed, relative accuracy, precision and efficiency) for use as part of quasi-automated ED-XRF spectrum evaluation procedure. To estimate successfully the net peak areas of small peaks, the Hypermet function is suitable to correct for the low-energy tailing of intense neighbouring peaks, so that, in combination with the LINEAR background method, more accurate results are obtained.

Regarding the quasi-unsupervised processing of large number of ED-XRF spectra constituting a μ -XRF image, we may conclude that it is necessary to use reli-

able spectrum evaluation procedures. Especially in the case of the use of synchrotron excitation, where considerable resources are employed to allow for the collection of XRF images, the use of the most accurate data handling during *a posteriori* processing appears mandatory in the case of trace element signals.

Acknowledgements

B.V. thanks the National Fund for Scientific Research (Brussels) for financial support; K.J. and P.V.E. are fellows of the same institution. The authors thank K. Jones (Brookhaven National Laboratories) for the use of the μ -XRF beamline X26A of the NSLS for the acquisition of the images shown in Fig. 6(a)–(c).

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